**CS 217: Practice questions on Linear regression and classification**

1. Suppose you have a regression task of predicting a real value in the interval [-4, 4] on a $d$-dimensional input $\mathbf{x} = [x_1, \ldots, x_d]$

   (a) First, assume we fit a model $f(\mathbf{x}) = \mathbf{w}^T\mathbf{x}+w_0$ on a training dataset $D = (\{(\mathbf{x}^1, y^1), \ldots, (\mathbf{x}^N, y^N)\}$ by minimizing the square loss. Let's say we know the optimal value of the $\mathbf{w}$ parameters. Call it $\mathbf{w}^*$. We only need to fit $w_0$ using the training dataset so as to minimize this loss. Provide a closed form expression for the fitted $w_0$?

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   (b) In the above if we change the loss to $\max(0, (y^i - f(\mathbf{x}^i))^2 - 1)$, and use gradient descent algorithm to find the optimal $w_0$, write the updated $w_0$ if initial $w_0 = 0$ and learning rate is $\eta$.

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   (c) Instead of using a linear regression function with the least square training objective, you convert this into a four class classification task as follows. First discretize [-4,4] into these four intervals

   $$[-4, -2), [-2, 0), [0, 2), [2, 4] \tag{1}$$

   Call each of these as class 1, 2, 3, 4 respectively. In the training data, any example with $y_i$ satisfying $-4 \leq y_i < -2$ will be assigned class 1, any with $y_i$ satisfying $-2 \leq y_i < 0$ will be assigned class 2, and so on. When the predicted class is "2" by the classifier, the predicted $y$ is midpoint "-1" of the corresponding interval [-2,0).

   Show a dataset where you expect the linear regression method to be better than the above classifier method, and another where the opposite is better. 1-dimensional dataset suffices.

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(d) Let $\mathbf{w}^y$ denote the parameters associated with class $y$. A limitation of training the classifier model with the default softmax loss below

$$\min_{\mathbf{w}^1,\mathbf{w}^2,\mathbf{w}^3,\mathbf{w}^4} \sum_{i=1}^{N} \mathbf{w}^{y_i}\mathbf{x}^i - \log \sum_{y} e^{\mathbf{w}^y \mathbf{x}^i}$$

is that the error function does not recognize that the four class labels are ordered. Ideally, the loss of misclassifying an example to a near-by class should be smaller than to a far-off class. Suggest a modification to the above loss function to fix this limitation.

2. Very often the training data for a classification task may assign wrong labels $y$ for an instance. One idea to fix the above problem is to make the loss function noise tolerant. Let $P_\theta(y|\mathbf{x})$ be the probability of some outcome that we want to maximize. The standard cross entropy loss minimizes the loss $L(\theta) = -\log P_\theta(y|\mathbf{x})$.

   (a) Plot $L(\theta)$ as a function of $P_\theta(y|\mathbf{x})$ on the x-axis.

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(b) Comment on why the above loss function is not a good idea when the label $y$ is noisy.

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(c) Suppose we modify the above loss to a generalized cross entropy loss defined as follows: $L_G(\theta) = -\log(P_\theta(y|\mathbf{x}) + 0.1)$. Plot $L_G(\theta)$ as a function of $P_\theta(y|\mathbf{x})$ on the x-axis.

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3. Suppose we wish to train a square error regression function with a $L\gamma$ regularizer. That is the training objective is made: $\min_{\mathbf{w}} \sum_i (y^i - \mathbf{w}.\mathbf{x}^i)^2 + C||\mathbf{w}||^\gamma$ where $||\mathbf{w}||^\gamma = \sum_j |w_j|^\gamma$. Suppose, we decide to use co-ordinate descent as follows:

$t = 0$, $\mathbf{w}^t = 0$
**while** change **do**
  **for** $j = 1 \ldots d$ **do**
    /*Minimize wrt $w_j$ keeping all other variables fixed at $\mathbf{w}^t$ */
    $w_j = \min_{w_j : w_k = w_k^t \forall k \neq j} \sum_i (y^i - \mathbf{w}.\mathbf{x}^i)^2 + C||\mathbf{w}||^\gamma$
    $\mathbf{w}^{t+1} = (w_1^t, \ldots w_{j-1}^t, w_j, w_{j+1}^t \ldots w_d^t)^T$
    $t = t + 1$
  **end for**
**end while**

(a) Solve for the optimum $w_j$ in step 4 for the case where $\gamma$ is even and $> 0$.

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(b) Assume that for every $C$ there exists a $C'$ such that the training objective is equivalent to

$$\min_{\mathbf{w}} \sum_i (y^i - \mathbf{w}.\mathbf{x}^i)^2 \\ \text{s.t. } ||\mathbf{w}||^\gamma < C' \tag{2}$$

Use this to write the optimal value $w_j$ in terms of $C'$ and $\mathbf{w}^t$-s when $\gamma = 1$.

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4. Consider the logistic classifier. If two features $f_k$ and $f_j$ are replicas of each other, state with a brief reason how the parameters $w_k$ and $w_j$ are related in the following two versions of the objectives used for training logistic classifiers: Assume the training data consists of instances of the form $(\mathbf{x}^i, y_i)$ where $y_i \in \{-1, 1\}$

(a) Unregularized: minimize $\sum_i \log(1 + e^{-\mathbf{w}^T\mathbf{x}^i y_i})$

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(b) Regularized with L2: minimize $C\sum_i \log(1 + e^{-\mathbf{w}^T\mathbf{x}^i y_i}) + \sum_j w_j^2$ where $C$ is a constant.

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(c) Regularized with L1: minimize $C\sum_i \log(1+e^{-\mathbf{w}^T\mathbf{x}^i y_i}) + \sum_j |w_j|$ where $C$ is a constant.

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5. In class we only considered square error for training a regression model. Propose an alternative error function and discuss pros and cons on account of accuracy, sensitivity to noisy training data, and training time.

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6. Let's say that we are trying to fit the parameters of a 3-way multinomial distribution using maximum likelihood estimation principle. We express the probability of the multinomial in terms of parameters $\{l_1, l_2, l_3\}$ as follows

$$f(x, \mathbf{w} = \{l_1, l_2, l_3\}) = \frac{e^{l_x}}{e^{l_1} + e^{l_2} + e^{l_3}}$$

For example, if $l_1 = 0, l_2 = 1.4, l_3 = 2$, then the probability of $x = 2$ is $\frac{e^{1.4}}{1+e^{1.4}+e^2}$

(a) Using the above form, write the log likelihood of the dataset $D = \{x^1, \ldots, x^N\}$ where each $x^i$ is either $1, 2$, or $3$.

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(b) Write the gradient of the above objective with respect to parameter $l_j$ for a training instance $x^i$.

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(c) If the initial value of the three parameters is $l_1 = 0, l_2 = 2, l_3 = -1$, for which value of $x^i$ is the norm of the gradient highest? Justify your answer.

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(d) If we obtain MLE estimate for the above model using the stochastic gradient descent algorithm with batch size one on a data set where outcome 3 is never observed, what is the smallest possible value of $l_3$ at the end of 100 gradient updates with a learning rate $\eta$ and initial value of $l_1 = l_2 = l_3 = 0$

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(e) Now say we redefine the probability using a smoothing constant $\epsilon$ as

$$f_\epsilon(x, \mathbf{w} = \{l_1, l_2, l_3\}) = (1 - 3\epsilon)\frac{e^{l_x}}{e^{l_1} + e^{l_2} + e^{l_3}} + \epsilon$$

Write the gradient of the maximum likelihood estimation objective of the above probability distribution with respect to parameter $l_2$ for a training example $x^i = 2$.

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(f) What are the pros and cons of using $f_\epsilon$ Vs $f$ for estimating density of a $K$ way categorical variable?

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7. Suppose we train a model for classification using the same square loss that is used for training regression models. Assume the labels are either +1 or -1. And our training objective is:

$$\min_{\mathbf{w}, w_0} \sum_{i=1}^{N} (y_i - (\mathbf{w}^T\mathbf{x} + w_0))^2 \tag{3}$$

Assume that the training data is separable by a linear decision boundary. Is the classifier trained using the above objective guaranteed to achieve zero error on the training data? Support your answer with proper justification.

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8. Let $f(x) = max(0, 1 - x)$ Find the subgradients of the function at 0, 1, and 2?
9. Questions 11.1 to 11.5 in Chapter 11.8 of PLS textbook.